University, an austere professor of physics came out of his study, waving around Einstein's article, screaming, 'The new Archimedes is born!'

But despite the outcry provoked by the step forwards made by Einstein in 1905, we are not yet at his masterpiece. Einstein's triumph is the *second* theory of relativity, the theory of *general relativity*, published ten years later, when he was thirty-five.

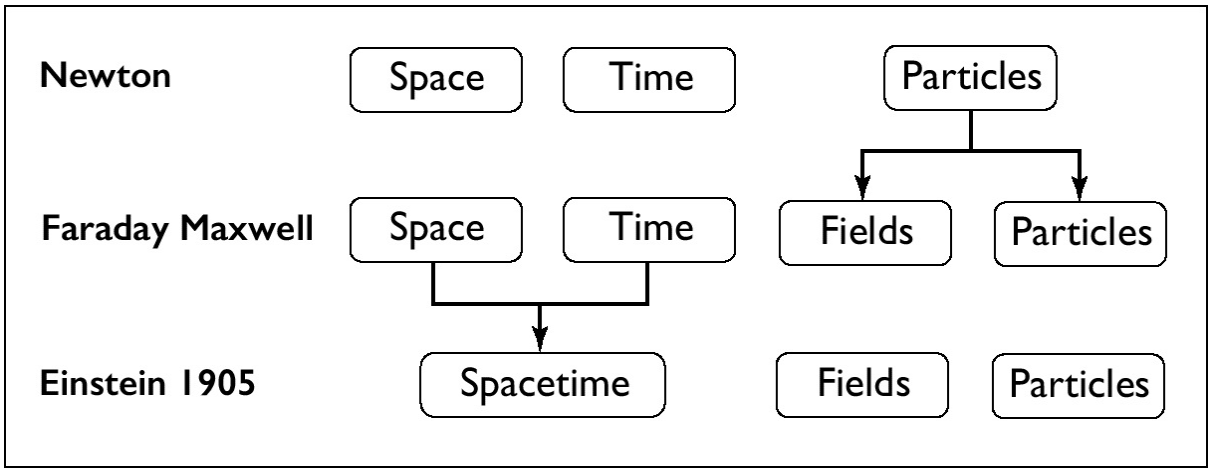


Figure 3.4 What is the world made of?

The theory of 'general relativity' is the most beautiful theory produced by physics, and the first of the pillars of quantum gravity. It is at the heart of the narrative of this book. Here, the real magic of twentieth-century physics begins.

The most beautiful of theories

After publishing the theory of special relativity, Einstein becomes a renowned physicist and receives offers of work from numerous universities. But something troubles him: special relativity does not square with what was known about gravity. He realizes this while writing a review on his theory, and wonders whether the venerable theory of the 'universal gravity' of the father of physics, Newton, should not be reconsidered as well, to make it compatible with his relativity.

The origin of the problem is easy to understand. Newton had tried to explain why things fall and planets revolve. He had imagined a 'force' that draws all bodies towards one another: the 'force of gravity'. How this force managed to draw distant things together without anything between them was not understood. Newton himself, as we have seen, had suspected that in the idea of a force acting between distant bodies that do not touch there was something missing; and that in order for the Earth to attract the Moon something that could transmit this force had to be there between the two. Two